

RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering

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MINI PROJECT REPORT

On

Stereo Vision Depth Estimation

using Camera Calibration and Depth from Focus

Algorithm: Depth from Focus (Laplacian Variance)

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1. Introduction

1.1 Background and Motivation

Depth estimation is a fundamental problem in computer vision with applications spanning autonomous vehicles, robotics, augmented reality, and 3D scene reconstruction. The ability to determine the distance of objects from a camera without specialized hardware has driven significant research in monocular and focus-based depth estimation techniques.

This mini project explores Depth from Focus (DFF) as a depth estimation approach. Unlike stereo vision — which requires two cameras separated by a known baseline — DFF estimates depth from a single camera by analysing sharpness variations across a sequence of images captured at different focal distances. This makes DFF more practical for scenarios where dual-camera setups are unavailable or impractical.

1.2 Problem Statement

Given a set of images of the same static scene captured from a fixed camera position, with each image focused at a different depth plane, the goal is to automatically estimate relative depth at every pixel using image sharpness as the primary cue. The output is a depth map in which colour encodes relative distance from the camera.

1.3 Objectives

- Implement a Depth from Focus pipeline using the Laplacian variance method.
- Design and apply a preprocessing pipeline that prepares raw images for accurate sharpness measurement.
- Generate a pixel-wise depth map from a focus stack of images.
- Visualize the resulting depth map using colour maps for intuitive interpretation.
- Evaluate the algorithm on self-captured image datasets under varied lighting conditions.

1.4 Scope of the Report

This report covers the complete implementation of the Depth from Focus algorithm using Python and OpenCV. It describes the dataset collection methodology, preprocessing steps, algorithm logic, and visual results. The report does not cover stereo disparity methods or calibrated metric depth reconstruction; it focuses on relative depth estimation from focus variation.

2. Literature Review

Depth from Focus (DFF) and Depth from Defocus (DFD) are classical passive depth estimation techniques studied since the late 1980s. This section reviews the key prior works relevant to the Laplacian variance approach implemented in this project.

2.1 Focus Measure Operators

Nayar and Nakagawa (1994) provided a comprehensive evaluation of focus measure operators for DFF. They categorised operators into spatial-domain and frequency-domain classes, finding that Laplacian-based operators offered an effective balance between noise sensitivity and computational cost. Subbarao et al. (1992) established the theoretical relationship between camera focus and depth via the Gaussian lens model, which underlies the DFF framework.

2.2 Laplacian Variance as a Sharpness Metric

The Laplacian operator computes second-order spatial derivatives of image intensity, thereby highlighting regions of rapid intensity change (edges). The variance of the Laplacian response serves as a focus measure: a sharp image contains strong high-frequency content reflected in high Laplacian variance, whereas a blurred image produces low variance. This metric was popularised for blur detection by Pech-Pacheco et al. (2000) and has since been widely adopted in focus stacking and DFF applications.

2.3 Deep Learning Approaches

Recent works, including Eigen et al. (2014) and Godard et al. (2017), have demonstrated that convolutional neural networks can learn depth from single monocular images, achieving metric depth estimation without explicit focus stacks. However, these methods require large labelled datasets and significant computational resources, making classical DFF approaches more accessible for resource-constrained academic projects.

2.4 Gap Identified

While deep-learning-based methods achieve higher accuracy, they lack transparency and interpretability in academic settings. Classical DFF using Laplacian variance provides a directly interpretable pipeline where each algorithmic stage can be understood, modified, and demonstrated — making it well-suited for an undergraduate mini project context.

3. Dataset Description

3.1 Self-Captured Focus Stack Dataset

For this project, images were captured using a mobile camera placed on a stable surface. A static scene was composed of 2-3 objects placed at varying distances from the camera (near, mid, and far planes). The camera was kept absolutely fixed between shots; only the focus point was varied by tapping different regions of the camera viewfinder.

This approach ensures that the fundamental assumption of DFF is satisfied: the same scene is observed from the same viewpoint, with only the focal plane changing across images.

3.2 Capture Protocol

- Camera position: Fixed on a flat surface (table), pointed at the scene.
- Objects used: Water bottle (near), book (middle), background wall or chair (far).
- Number of images: 5-8 images per dataset.
- Resolution: Native mobile camera resolution (transferred and stored as JPEG).
- Lighting: Indoor ambient lighting, kept constant throughout capture.

Table 1: Dataset Summary

Property	Details
Number of images (Dataset 1)	5
Number of images (Dataset 2)	8
Image format	JPEG (.jpg)
Focus levels	Near, Mid, Far (and intermediate)
Camera movement	None
Scene type	Indoor, static objects
Depth variation	Approx. 30 cm to 150 cm from lens

4. Preprocessing Methodology

A preprocessing pipeline is applied uniformly to all images in the focus stack before the Laplacian computation. The pipeline ensures consistent image quality and reduces artefacts that could corrupt the focus measure.

4.1 Image Loading and Format Conversion

All images are read using OpenCV's `cv2.imread()` function, which loads images in BGR channel order. File names are sorted to maintain a consistent ordering of the focus stack. Non-image files are filtered out to prevent loading errors.

```
import cv2, os
image_folder = 'images'
filenames = sorted(os.listdir(image_folder))
images = []
for file in filenames:
    if file.endswith(('.jpg', '.png', '.jpeg')):
        img = cv2.imread(os.path.join(image_folder, file))
        if img is not None:
            images.append(img)
```

4.2 Grayscale Conversion

Each colour image is converted to grayscale before applying the Laplacian operator. This is essential because the Laplacian focus measure is defined on a single-channel intensity image. Colour information is not used in the sharpness computation.

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

4.3 Gaussian Denoising (Optional Enhancement)

A Gaussian blur with a small kernel can be optionally applied before the Laplacian to reduce sensor noise, which would otherwise artificially inflate the sharpness score. A 3x3 or 5x5 kernel is used with a small sigma value to preserve edge information.

```
gray_denoised = cv2.GaussianBlur(gray, (3, 3), 0.5)
```

4.4 Resolution Standardisation

If images in the focus stack have different resolutions (which can occur when different camera modes are used), all images are resized to a common resolution. Mismatched resolutions would cause the NumPy stack operation to fail.

```
img_resized = cv2.resize(img, (target_width, target_height))
```

5. Algorithm: Depth from Focus (Laplacian Variance)

5.1 Algorithm Overview

The Depth from Focus algorithm estimates the relative depth of each pixel by determining which image in the focus stack renders that pixel most sharply. The core principle is: an object appears sharpest in the image where the camera focal plane coincides with the object's depth. By finding the focus stack index that yields maximum sharpness at each pixel, the algorithm constructs a depth map.

5.2 The Laplacian Operator

The Laplacian is a second-order derivative operator that computes the sum of second partial derivatives of image intensity in the x and y directions. It is sensitive to rapid intensity transitions (edges and textures), which are the defining characteristics of sharp, in-focus regions.

Mathematically, the discrete Laplacian at pixel (x, y) is:

$$L(x,y) = I(x+1,y) + I(x-1,y) + I(x,y+1) + I(x,y-1) - 4 \times I(x,y)$$

where $I(x,y)$ denotes the grayscale intensity at pixel (x,y). A high absolute Laplacian value at a pixel indicates a sharp edge (in focus); a low value indicates a smooth, blurred region (out of focus).

5.3 Pixel-wise Depth Computation

For each pixel (x,y) in the image, the algorithm selects the focus stack index k that maximises the absolute Laplacian response:

$$\text{depth}(x,y) = \text{argmax}_k |L_k(x,y)|$$

This index is interpreted as a relative depth label: lower indices (images focused near) assign depth to near objects; higher indices assign depth to far objects.

5.4 Implementation Steps

1. Load all images from the focus stack into a list.
2. Convert each image to grayscale.
3. Apply the Laplacian operator to each grayscale image using `cv2.Laplacian(gray, cv2.CV_64F)`.
4. Take the absolute value of each Laplacian response map.
5. Stack all Laplacian maps into a 3D NumPy array (height x width x num_images).
6. Use `np.argmax()` along the depth axis to select the sharpest image index at each pixel.
7. Normalize the resulting depth map to the 0-255 range.
8. Apply a colour map (`COLORMAP_JET`) and display the result.

5.5 Complete Source Code

```
import cv2
import numpy as np
import os
import matplotlib.pyplot as plt

# ----- Step 1: Load Images
image_folder = 'images'
images = []
filenames = sorted(os.listdir(image_folder))

for file in filenames:
    if file.endswith(('.jpg', '.png', '.jpeg')):
        path = os.path.join(image_folder, file)
        img = cv2.imread(path)
        if img is not None:
            images.append(img)
            print(f'Loaded: {file}')

print(f'Total images: {len(images)}')

# ----- Step 2: Convert to Grayscale -----
gray_images = [cv2.cvtColor(img, cv2.COLOR_BGR2GRAY) for img in images]

# ----- Step 3: Compute Laplacian -----
laplacians = [cv2.Laplacian(gray, cv2.CV_64F) for gray in gray_images]
laplacians = [np.abs(lap) for lap in laplacians]

# ----- Step 4: Stack and Compute Depth -----
lap_stack = np.stack(laplacians, axis=-1)
depth_map = np.argmax(lap_stack, axis=-1)
print('Depth map created!')

# ----- Step 5: Normalize -----
depth_normalized = cv2.normalize(depth_map, None, 0, 255, cv2.NORM_MINMAX)
depth_normalized = depth_normalized.astype('uint8')

# ----- Step 6: Apply Color Map -----
depth_colored = cv2.applyColorMap(depth_normalized, cv2.COLORMAP_JET)
depth_rgb = cv2.cvtColor(depth_colored, cv2.COLOR_BGR2RGB)

# ----- Step 7: Display -----
plt.imshow(depth_rgb)
plt.title('Depth Map using Laplacian Focus')
plt.axis('off')
plt.show()
```

5.6 Algorithm Flowchart

[Note: Insert a flowchart diagram here. Suggested tool: draw.io or Lucidchart. The flow should include the stages below.]

Table 2: Algorithm Pipeline Stages

Stage	Operation	OpenCV / NumPy Function
1. Image Loading	Read all JPEG/PNG files from folder	cv2.imread()

Stage	Operation	OpenCV / NumPy Function
2. Grayscale Conversion	Convert BGR to single-channel intensity	cv2.cvtColor()
3. Laplacian Filter	Apply second-order derivative operator	cv2.Laplacian(gray, CV_64F)
4. Absolute Value	Convert to positive sharpness map	np.abs()
5. Stack	Create 3D array of all Laplacian maps	np.stack(axis=-1)
6. Argmax	Find sharpest image index per pixel	np.argmax(axis=-1)
7. Normalise	Scale depth values to 0-255	cv2.normalize()
8. Colormap	Apply JET colour map for visualisation	cv2.applyColorMap()

6. Results and Analysis

6.1 Qualitative Results

The depth map generated by the algorithm displays distinct colour regions corresponding to objects at different distances from the camera. Warmer colours (red, orange) indicate pixels that were sharpest in images focused on near objects, while cooler colours (blue, green) indicate regions that were sharpest in images focused on distant objects.

[Note: Insert screenshots here — Sample Image, Depth Map output, and side-by-side comparison.]

6.2 Effect of Number of Images

The number of images in the focus stack directly affects the resolution of depth levels in the output. With 2 images, the depth map is binary (near or far). With 5-8 images, intermediate depth levels are captured, resulting in a smoother, more detailed depth map.

Table 3: Effect of Focus Stack Size

Number of Images	Depth Levels	Map Quality	Processing Time
2	2 (binary)	Rough	Fast
4	4	Moderate	Moderate
6-8	6-8	Smooth gradient	Moderate
10+	10+	High detail	Slower

6.3 Observations

- Fixed camera position is critical — any camera movement between shots introduces misalignment and degrades depth map quality.
- The Laplacian response is sensitive to noise; applying a mild Gaussian blur before the Laplacian reduces false high-frequency responses.
- Objects with uniform textures (e.g., plain walls) produce low Laplacian values across all images, making depth assignment ambiguous for those regions.
- The depth map provides relative depth ordering, not absolute metric distance. Objects can be ranked as near/mid/far but actual distances in metres cannot be recovered without camera calibration.

7. Discussion

7.1 Strengths of the Approach

The Depth from Focus method using Laplacian variance offers several practical advantages for an academic mini project. The algorithm is transparent and interpretable: each stage (grayscale conversion, Laplacian, argmax) has a clear physical motivation that can be explained and demonstrated. The implementation requires only a standard laptop with Python and OpenCV, with no specialised hardware beyond a mobile camera.

The use of a mobile camera for image capture is particularly effective because modern smartphone cameras allow the user to tap to set the focal point, providing precise control over which depth plane is in focus in each shot.

7.2 Limitations

The primary limitation of the DFF approach is the requirement for multiple images of the same static scene. Moving objects between shots, or any camera movement, violates the fixed-viewpoint assumption and produces erroneous depth estimates. The method also produces only relative depth and cannot determine the actual distance of objects in metres without additional camera calibration.

Textureless surfaces and uniform colour regions produce weak Laplacian responses across all images, making the depth assignment unreliable in those areas. Additionally, the discrete focus stack introduces quantisation in the depth levels: the finest depth resolution is limited by the number of images captured.

7.3 Comparison with Stereo Vision

Table 4: Depth from Focus vs Stereo Vision

Aspect	Depth from Focus (This Project)	Stereo Vision
Cameras required	1 (single camera)	2 (stereo pair)
Depth cue used	Focus / blur variation	Disparity between views
Setup complexity	Low	High (calibration needed)
Output	Relative depth map	Metric depth map
Main limitation	Requires static scene; multiple shots	Requires calibrated rig; occlusions
Best use case	Macro photography, controlled scenes	Robotics, autonomous vehicles

8. Conclusion and Future Work

8.1 Conclusion

This mini project successfully implemented the Depth from Focus (Laplacian Variance) algorithm for relative depth estimation from a focus stack of images. The system was built from scratch using Python, OpenCV, NumPy, and Matplotlib. Images were captured using a mobile camera under controlled conditions, and the algorithm produced a meaningful depth map that correctly distinguished near and far objects in the scene.

The project demonstrated a complete image processing pipeline: from raw image acquisition and preprocessing to sharpness measurement, depth map generation, and colour visualisation. The key contribution is a working, interpretable implementation that can be replicated with minimal hardware requirements.

8.2 Future Work

- Extend the dataset to include more images (8-10 per stack) for smoother multi-level depth maps.
- Integrate camera calibration parameters to convert relative depth into approximate metric distances.
- Compare Laplacian variance with other focus measures such as Tenengrad, variance of image gradient, or wavelet-based operators.
- Implement a focus stack merge (all-in-focus image) alongside the depth map as an additional output.
- Deploy the pipeline on a Raspberry Pi to evaluate real-time performance on edge hardware.
- Extend towards a hybrid approach combining Depth from Focus with stereo disparity for improved accuracy.

References

[References in IEEE format. Minimum 6 references required.]

- [1] S. K. Nayar and Y. Nakagawa, "Shape from focus," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 16, no. 8, pp. 824-831, Aug. 1994.
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- [6] G. Bradski and A. Kaehler, Learning OpenCV: Computer Vision with the OpenCV Library. O'Reilly Media, 2008.
- [7] OpenCV Documentation, "cv2.Laplacian," Available: <https://docs.opencv.org/4.x/>
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Appendix A — Viva Preparation

The following questions are commonly asked in mini project vivas. Study the answers carefully.

Q1: What is Depth from Focus?

Depth from Focus is a technique that estimates the relative depth (distance) of objects in a scene by analysing how sharpness varies across a sequence of images captured with different focal settings. Objects appear sharpest in the image where the camera's focal plane coincides with their distance.

Q2: Why is the Laplacian operator used?

The Laplacian operator computes second-order spatial derivatives of image intensity, making it sensitive to edges and rapid intensity changes. Sharp, in-focus regions contain strong high-frequency content reflected in high Laplacian variance, whereas blurred, out-of-focus regions produce low Laplacian values. It is computationally efficient and easy to implement using OpenCV.

Q3: Why must the camera remain fixed?

The DFF algorithm compares pixel values at the same (x, y) coordinate across all images. If the camera moves, the same pixel location in different images corresponds to different physical points in the scene, making the sharpness comparison meaningless and corrupting the depth map.

Q4: What does the colour in the depth map represent?

The colours represent relative depth levels, not absolute metric distances. Warmer colours (red) indicate pixels that were most in focus in images taken with the near focal setting (closer objects), while cooler colours (blue) indicate pixels focused in images taken with a far focal setting (distant objects). The colour is a visual encoding of the focus stack index that yielded maximum sharpness.

Q5: What is the difference between Depth from Focus and Stereo Vision?

Depth from Focus uses a single camera and analyses blur variation across multiple images taken at different focal distances. Stereo vision uses two cameras separated by a known baseline and estimates depth from the pixel displacement (disparity) between the two views. DFF requires a static scene with multiple shots; stereo vision can estimate depth from a single pair of simultaneous images but requires a calibrated stereo rig.

Q6: What is the limitation of this method?

The main limitations are: (1) it requires multiple images of a static scene (dynamic scenes cannot

be processed); (2) it provides only relative depth, not metric distances; (3) uniform or textureless

regions produce weak Laplacian responses, leading to unreliable depth estimates in those areas;
(4) the depth resolution is limited by the number of images in the focus stack.

Q7: Which line in the code applies the Laplacian?

```
laplacians = [cv2.Laplacian(gray, cv2.CV_64F) for gray in gray_images]
```

This line applies the Laplacian operator to every grayscale image in the focus stack and stores the resulting maps in a list.

Appendix B — Project Setup and Running Guide

B.1 Environment Requirements

Component	Version / Details
Python	3.10 or 3.11
OpenCV	pip install opencv-python
NumPy	pip install numpy
Matplotlib	pip install matplotlib
IDE	Visual Studio Code (recommended)
OS	Windows 10/11, macOS, or Linux

B.2 Folder Structure

```
StereoDepthProject/  
  main.py  
  images/  
    img1.jpg  
    img2.jpg  
    img3.jpg  
    img4.jpg  
    img5.jpg
```

B.3 Image Capture Protocol

- Place 2-3 objects at different distances on a table (near, mid, far).
- Rest the mobile camera on a stable surface, pointed at all objects.
- Without moving the camera, tap the nearest object to focus on it. Capture.
- Tap the middle object to change focus. Capture.
- Tap the farthest object. Capture.
- Take 2-3 more shots with intermediate focus points.
- Transfer all images to the laptop and place in the images/ folder.
- Rename files as img1.jpg, img2.jpg, ... in focus order (near to far).

B.4 Running the Project

```
# 1. Open terminal in project folder  
# 2. Install dependencies  
pip install opencv-python numpy matplotlib  
  
# 3. Run the project  
python main.py
```

B.5 Sharing the Project

- Right-click the StereoDepthProject folder.
- Click Send to > Compressed (zipped) folder.

19. Send StereoDepthProject.zip via WhatsApp Web, Google Drive, or email.
20. Recipient extracts the ZIP, installs libraries, and runs python main.py.